STATISTICAL ANALYSIS AND PERSONALIZED BOOK RECOMMENDATION SYSTEM BASED ON USER PREFERENCE DATA

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Introduction

- The rapid growth of digital book platforms has made it difficult for users to find books that match their interests.
- Traditional search and recommendation systems are time-consuming and less personalized .
- However, existing systems face issues like **cold start (no data), data sparsity (less data), and limited personalization**. These problems reduce the accuracy and effectiveness of recommendations.
- This project proposes a **Statistical Analysis and Personalized Book Recommendation System** to overcome these challenges.
- This project develops a statistical and hybrid recommendation system to provide accurate and personalized book suggestions.

Motivation for Research

- Existing book recommendation systems provide less accurate results due to limited techniques and poor use of data.
- There is a need to use statistical analysis to better understand user behavior and improve prediction accuracy.
- Statistical methods work by analyzing rating patterns, similarities, and trends to generate reliable recommendations.
- Traditional systems do not combine methods effectively, so a hybrid approach can improve performance.
- This research aims to develop a more accurate, scalable, and personalized recommendation system for better user experience.

Literature Review

Paper Title	Author	Approach	Findings	Limitations
Web-Based Book Recommendation System using Collaborative Filtering	Mankar et al. (2023)	Collaborative filtering (user-item similarity), web-based system	Recommends books based on similar users' ratings and preferences	Cold start problem; data sparsity; no hybrid or content-based improvement
Tag-Based Book Recommendation using Data Mining	Wang & Wang (2024)	Tag-based recommendation with data mining	Uses user-generated tags to improve personalization and relevance	Depends on tag availability; noisy/inconsistent tags; weak statistical modeling
Book Recommendation using WALS Approach	Kavitha & Murugesan (2024)	Weighted ALS, matrix factorization	High prediction accuracy; works well for large datasets	Needs large data; computationally expensive; low explainability

Paper Title	Author	Approach	Findings	Limitations
Recommendation Systems: A Systematic Review	Malakul & Songmuang (2026)	Survey and comparative analysis	Covers latest techniques like AI and hybrid systems	No implementation; no new model; limited validation
Hybrid/Advanced Recommendation Insights (from gaps across studies)	Derived from literature gaps	Combination of CF, content-based, and statistical methods	Shows need for hybrid and data-driven approaches for better accuracy	Existing works do not fully integrate statistical + hybrid methods effectively

Problem Statement

- Existing recommendation systems rely on single methods, resulting in limited accuracy and performance.
- Techniques like collaborative filtering face cold start and data sparsity issues, reducing effectiveness.
- Tag-based and content-based methods depend on data quality, which may be inconsistent or incomplete.
- Advanced models require high computational cost and large datasets, making them less efficient.
- There is a need for a hybrid, statistically driven system that improves accuracy, scalability, and personalization.

Research Objectives

Objective 1 :To Develop a Hybrid Recommendation System
Developing a system that combines collaborative and content-based filtering to generate better and more reliable book recommendations than single-method systems.

Objective 2 : To Handle Cold Start and Data Sparsity Problems
Handling situations with limited or no data by using statistical methods and user preferences to generate effective recommendations for new users and books.

Objective 3 : Enhancing Personalization in Book Recommendations Using User Preference Data
Giving the right recommendations to each individual user based on their preferences, interests, and past behavior.

Objective 4 : To Improve Scalability and Computational Efficiency
Improving system performance by applying efficient algorithms to handle large datasets quickly and generate faster results with minimal time and resources..

Objective 5: To Improve Recommendation Quality and Explainability
Improving recommendation accuracy and explaining why each book is suggested, making the system more reliable and user-friendly.

Methodology

Dataset Collection:
Book recommendation dataset from Kaggle with over 1 million samples and 6–8 variables including user and book details.

Data Preprocessing:
Cleaning data by handling missing values, removing duplicates, and converting it into a structured format.

Exploratory Data Analysis:
Analyzing user behavior, rating patterns, and identifying popular books and genres.

Statistical Analysis:
Applying similarity measures like cosine and correlation to find relationships between users and books.

Recommendation System:

Using collaborative filtering, content-based filtering, and a hybrid approach to built the recommendations system

Result Generation:

Predicting user preferences and providing Top-N personalized book recommendations.

Evaluation:

The system generates personalized book recommendations and is evaluated using performance metrics like accuracy and RMSE(Root Mean Square Error).

References

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THANK YOU

Queries & Suggestions